Communication

Advancements in Drug Repurposing: In-Silico Analysis of Sildenafil

Phoenix Blueprint Subsystem
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Abstract: Sildenafil, a phosphodiesterase-5 (PDE5) inhibitor, has shown potential for repurposing beyond its original indication for erectile dysfunction. Its mechanism of action, increasing blood flow by enhancing nitric oxide (NO) levels, suggests potential applications in conditions characterized by impaired blood flow or vascular dysfunction. Clinical trials and literature reviews indicate promise in areas such as pulmonary arterial hypertension and possibly in the treatment of ischemic stroke, given its effects on vascular function.

Keywords: Sildenafil; drug repurposing; Phoenix Score; computational prediction

1 1. Introduction

A drug candidate discovered in the research phase has an approximately 10% likelihood of gaining regulatory approval following clinical trials. This is partly due to the difficulty in accurately predicting the efficacy and safety of drugs in humans from non-clinical tests. Drug repurposing, which involves utilizing already approved drugs or clinically developed compounds, has been widely practiced to mitigate these risks.

This study aimed at uncovering recent trends in drug repurposing approaches and evaluating the potential of **Sildenafil**. Utilizing the Phoenix Scoring Framework, we generated an objective viability score to quantify the compound's likelihood of successful redeployment into novel therapeutic indications.

2 2. Results

2.1 2.1. Market Intelligence and Opportunities

The repurposing opportunities for Sildenafil were identified by traversing disease networks. The following table outlines the top repurposing opportunities, prioritized by clinical phase proximity and addressable market size.

Indication	**Max Phase**	**TAM**	**Score**
Type 2 Diabetes Mellitus	PHASE3	$64.4B	7.5
Erectile Dysfunction	PHASE4	$4.5B	5.9
Endometrial Preparation	PHASE4	$0.5B	5.5
Rectal Tumors	PHASE4	$0.2B	5.4
Familial Persistent, Of The Newborn Pulmonary Hypertension	PHASE3	$0.1B	4.6

2.2 2.2. Clinical Trial Evidence

To accurately estimate the efficacy and safety of Sildenafil, extrapolation from clinical trials to novel applications is critically required. Analysis of clinical trial registries highlights the following studies which underpin our target predictions:

- **Unknown ID** (EARLY_PHASE1): Testosterone for Penile Rehabilitation After Radical Prostatectomy. Status: *TERMINATED*
- **Unknown ID** (PHASE1): Sildenafil (Viagra) Treatment of Subacute Ischemic Stroke. Status: *TERMINATED*
- **Unknown ID** (PHASE1): Evaluation of Metabolic Markers for the Prediction of DDI of Various CYP3A Substrates and Inhibitors. Status: *COMPLETED*
- **Unknown ID** (PHASE3): Acute Effects of Sildenafil on Endothelial Function in People With Diabetes. Status: *COMPLETED*
- **Unknown ID** (PHASE1): Sildenafil 100 mg Tablets Relative to Viagra 100 mg Tablets. Status: *UNKNOWN*

2.3 2.3. Intellectual Property Landscape

Drug repurposing relies heavily on evaluating the patent landscape to formulate a clear commercialization strategy. The major linked patents retrieved for Sildenafil are:

- **US5250534**: Patent US5250534
- **US6469012**: Patent US6469012
- **US11337979**: Patent US11337979

3 3. Discussion

Based on the aggregated metrics natively processed via the pipeline, **Sildenafil** exhibits a **Phoenix Score of 9.0/10**.

This highlights that drug repurposing in this compound is not limited to the conventional searches but could be aimed at alternative development concepts. It is expected that sophisticated drug repurposing approaches combined with algorithmic evaluations will accelerate the translation into clinical trials.